rainbowing making artificial ISlands (like eggburg amsterdam)

5 August 2026

Rainbowing is a high-efficiency dredging technique where a massive ship shoots a mixture of sand and water through the air in a giant arc to build new land. The Dutch have mastered this process to create the IJburg archipelago in Amsterdam (often phonetically searched as "eggburg"), an ongoing multi-decade expansion project in Lake IJmeer consisting of multiple custom-built artificial islands. [1, 2, 3, 4, 5]


I'm an Amsterdam local

How Rainbowing Works

Instead of using slow conveyor belts or traditional underwater pipelines, maritime construction crews use specialized vessels called **Trailing Suction Hopper Dredgers (TSHD)**. [1, 2]

1. **Sucking up Sand**: The ship sails to a designated area, dragging huge vacuum-like pipes along the sea or lake bed to fill its internal hold (the hopper) with sand. [1]
2. **Mixing the Slurry**: The sand is mixed with high-pressure water inside the ship to create a thick, liquid mud slurry. [1]
3. **The "Rainbow" Launch**: Powerful onboard pumps blast this slurry out of a specialized bow nozzle into the air. It flies in a massive, colorful arc directly into the shallow target zone. [1, 2, 3]
4. **Natural Settling**: Landing through the air allows the water to drain away quickly, leaving behind dense, tightly packed sand that is instantly stable enough to build on. [1, 2]

The IJburg "Pancake Method"

In Amsterdam's delicate lake ecosystem, engineers combined rainbowing with what they call the **"pancake method"** to build islands like Haveneiland and Strandeiland. [1, 2, 3]

- **Porous Screens**: Large mesh fabric screens are placed underwater to mark the borders of the future island.

- **Layering**: Dredgers spray a thin layer of sand inside the boundaries, acting like pancake batter.
- **Consolidation**: Crew members let the layer sit and settle so the water drains through the mesh.
- **Stacking**: They repeat the spraying process layer by layer until the artificial island reaches roughly **2 meters (6.5 feet) above the water line**. [1, 2]

By using this step-by-step approach instead of dumping massive amounts of earth all at once, engineers protect local wildlife from dangerous silt buildup and prevent the island's foundation from shifting. Today, these recycled sandbars support eco-friendly neighborhoods featuring modern tram lines, apartment complexes, and floating homes. [1, 2, 3, 4]

If you are researching maritime engineering, I can provide information on **how engineers protect these islands from storms**, or outline the **environmental impacts** of dredging. Which angle should we explore next? [1]